

Math Thermo  
class # 05  
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## Legendre Transforms and Thermodynamics Potentials

Defn (5.1): Suppose that  $f: [a, b] \rightarrow \mathbb{R}$  is continuous. For every  $p \in [c, d]$ , define

$$(5.1) \quad f^*(p) := \sup_{x \in [a, b]} \{xp - f(x)\}$$

The function  $f^*: [c, d] \rightarrow \mathbb{R}$  is called the Legendre transform of  $f$ .

Theorem (5.2): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$  with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

Then, for all  $p \in \text{Range}(f')$ ,

$$(5.2) \quad f^*(p) = x_p \cdot p - f(x_p),$$

where  $x_p \in [0, \infty)$  is the unique solution to

$$f'(x_p) = p.$$

Proof: let  $p \in \text{Range}(f')$ .  $f'([0, \infty)) \xrightarrow{\text{onto}} \text{Range}(f')$ .  
For every  $p \in \text{Range}(f') \exists! x_p \in [0, \infty)$  such that,

$$f'(x_p) = p,$$

Since  $f': [0, \infty)$  is strictly increasing.

Fix  $p \in \text{Range}(f')$ . By Taylor's Theorem,

$$f(x) = f(x_p) + p(x - x_p) + \frac{1}{2}f''(\xi_p)(x - x_p)^2$$

for some  $\xi_p$  between  $x$  and  $x_p$ .

$$\begin{aligned} \sup_{0 \leq x < \infty} \{x \cdot p - f(x)\} \\ &= \sup_{0 \leq x < \infty} \{x_p \cdot p - f(x_p) - \frac{1}{2}f''(\xi_p)(x - x_p)^2\} \\ &= x_p \cdot p - f(x_p). \quad /// \end{aligned}$$

Theorem (5.3): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$   
with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

$$(5.3) \quad \text{Then, } [f^*]'(p) = [f']^{-1}(p), \quad [f^*]''(p) > 0,$$

for all  $p \in \text{Range}(f')$ .

Proof: Recall

$$f': [0, \infty) \xrightarrow[\text{onto}]{1-1} \text{Range}(f')$$

Thus

$$f'([f']^{-1}(p)) = p, \quad \forall p \in \text{Range}(f').$$

Also, observe that  $\frac{d}{dp}([f']^{-1}) \in C(\text{Range}(f'); \mathbb{R})$ .

Now,

$$\begin{aligned}
 \frac{d}{dp} [f^*(p)] &= \frac{d}{dp} \{x_p \cdot p - f(x_p)\} \\
 &= \frac{d}{dp} \{[f']^{-1}(p) \cdot p - f([f']^{-1}(p))\} \\
 &= p \cdot \frac{d}{dp} [f']^{-1}(p) \\
 &\quad + [f']^{-1}(p) \\
 &\quad - \underbrace{\frac{df}{dx}([f']^{-1}(p))}_{=p} \cdot \frac{d}{dp} [f']^{-1}(p) \\
 &= [f']^{-1}(p).
 \end{aligned}$$

Now that this is established, we have

$$\frac{d^2}{dp^2} [f^*(p)] = \frac{d}{dp} [f']^{-1}(p)$$

for all  $p \in \text{Range}(f')$ . Since  $f'' \in C([0, \infty))$  and

$$f''(x) > 0, \quad \forall x \in [0, \infty),$$

$$\frac{d}{dp} [f']^{-1}(p) > 0, \quad \forall p \in \text{Range}(f'). \quad \text{///}$$

Finally, we observe the following.

Theorem (5.4): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$   
with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

Then  $f^* \in C^2(\text{Range}(f'); \mathbb{R})$  with

$$[f^*]''(p) > 0, \quad \forall p \in \text{Range}(f').$$

Furthermore

$$(5.3) \quad [f^*]^*(x) = f(x), \quad \forall x \in [0, \infty).$$

In other words, the Legendre transform is involutory  
and information preserving.

Proof: It follows that, for all  $x \in [0, \infty)$ ,

$$[f^*]^*(x) = x p_x - f^*(p_x),$$

where  $p_x \in \text{Range}(f')$  is the unique solution to

$$f^{*'}(p_x) = x.$$

Recall

$$f^*(p_x) = p_x \cdot x_{p_x} - f(x_{p_x}),$$

where  $x_{p_x} \in [0, \infty)$  is the unique solution to

$$f'(x_{p_x}) = p_x.$$

Of course, by uniqueness,

$$x_{p_x} = x.$$

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$$\begin{aligned} [f^*]^*(x) &= x p_x - f^*(p_x) \\ &= x \cdot p_x - \{ p_x \cdot x p_x - f(x p_x) \} \\ &= f(x). \quad /// \end{aligned}$$

Why is the Legendre transform useful in Thermodynamics?

This transform allows us to introduce new thermodynamic coordinates/variables.

Recall

$$\tilde{U} = \tilde{U}(S, V, N)$$

or

$$d\tilde{U} = T dS - p dV + \mu dN$$

The latter means

$$T = \frac{\partial \tilde{U}}{\partial S} \quad -p = \frac{\partial \tilde{U}}{\partial V} \quad \mu = \frac{\partial \tilde{U}}{\partial N}$$

in short hand

| Extensive Quantity | Conjugate Variable | Variable |
|--------------------|--------------------|----------|
| $\tilde{U}$        | $T$                | $S$      |
|                    | $-p$               | $V$      |
|                    | $\mu$              | $N$      |
| $\tilde{S}$        | $1/T$              | $U$      |
|                    | $p/T$              | $V$      |
|                    | $-\mu/T$           | $N$      |

Definition (5.5): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary ( $r=1$ ). Define

$$(5.4) \quad \tilde{F}(T, V, N) = \tilde{U}(S_T, V, N) - TS_T$$

where  $S_T = S_T(V, N)$  is the unique solution of the equation

$$(5.5) \quad T_U(S_T(V, N), V, N) = T$$

for fixed values of  $N$  and  $V$ .

$\tilde{F}$  is called the Helmholtz free energy.

Formally, we have

$$d\tilde{F} = d\tilde{U} - SdT - TdS$$

$$= TdS - pdV + \mu dN - SdT - TdS$$

$$= -SdT - pdV + \mu dN.$$

| Extensive Quantity | Conjugate Variable | Variable |
|--------------------|--------------------|----------|
| $\tilde{U}$        | $T$                | $S$      |
|                    | $-p$               | $V$      |
|                    | $\mu$              | $N$      |
| $\tilde{F}$        | $-S$               | $T$      |
|                    | $-p$               | $V$      |
|                    | $\mu$              | $N$      |

Example (5.6): Suppose that

$$\tilde{S}(U, V, N) = \left( \frac{N V U R^2}{v_0 \theta} \right)^{1/3}.$$

It follows that

$$\tilde{U}(S, V, N) = \frac{S^3 v_0 \theta}{N V R^2}.$$

The temperature function is

$$T_U(S, V, N) = \frac{3 S^2 v_0 \theta}{N V R^2}.$$

Or

$$\begin{aligned} \frac{1}{T_S(U, V, N)} &= \frac{1}{3} \left( \frac{N V U R^2}{v_0 \theta} \right)^{-2/3} \frac{N V R^2}{v_0 \theta} \\ &= \frac{1}{3} \left( \frac{N V R^2}{v_0 \theta} \right)^{1/3} U^{-2/3}. \end{aligned}$$

Thus

$$T_S(U, V, N) = 3 \left( \frac{v_0 \theta}{N V R^2} \right)^{1/3} U^{2/3}.$$

Recall that

$$T_U(S, V, \vec{N}) = T_S(\tilde{U}(S, V, N), V, \vec{N})$$

$$T_S(U, V, \vec{N}) = T_U(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

By definition

$$\tilde{F}(T, V, N) = \tilde{U}(S_T, V, N) - T S_T$$

where

$$T_U(S_T(V, N), V, N) = T.$$

For our example,

$$T_U(S, V, N) = \frac{3 S^2 v_0 \theta}{N V R^2}.$$

Hence,

$$S_T = \sqrt{\frac{T N V R^2}{3 v_0 \theta}}.$$

Thus,

$$\begin{aligned} \tilde{F}(T, V, N) &= \tilde{U} \left( \sqrt{\frac{T N V R^2}{3 v_0 \theta}}, V, N \right) \\ &\quad - T \sqrt{\frac{T N V R^2}{3 v_0 \theta}} \end{aligned}$$

But

$$\begin{aligned} \tilde{U}(S_T, V, N) &= \frac{S_T^3 v_0 \theta}{N V R^2} \\ &= \left( \frac{T N V R^2}{3 v_0 \theta} \right)^{3/2} \frac{v_0 \theta}{N V R^2} \end{aligned}$$

So that

$$\begin{aligned} \tilde{F}(T, N, V) &= \left( \frac{T N V R^2}{3 v_0 \theta} \right)^{3/2} \frac{v_0 \theta}{N V R^2} \\ &\quad - T \sqrt{\frac{T N V R^2}{3 v_0 \theta}} \end{aligned}$$



$$= \frac{1}{3} \frac{T^{3/2}}{\sqrt{3}} \sqrt{\frac{NVR^2}{v_0 \theta}} - \frac{T^{3/2}}{\sqrt{3}} \sqrt{\frac{NVR^2}{v_0 \theta}}$$

Finally,

$$\tilde{F}(T, N, V) = -\frac{2T^{3/2}}{3} \sqrt{\frac{NVR^2}{3v_0 \theta}}.$$

Now,

$$\frac{\partial \tilde{F}}{\partial T} = -\frac{2}{3} \cdot \frac{3}{2} \cdot \sqrt{T} \cdot \sqrt{\frac{NVR^2}{3v_0 \theta}}.$$

But

$$T_U(S, V, N) = \frac{3 S^2 v_0 \theta}{NVR^2}$$

Thus,

$$\sqrt{T} = \sqrt{\frac{3v_0 \theta}{NVR^2}} S_T$$

and

$$\frac{\partial \tilde{F}}{\partial T} = -S_T.$$

Theorem (5.7): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary ( $r=1$ ). The following provides an equivalent means for finding the Helmholtz free energy:

$$\tilde{F}(T, V, N) = U_T - T \tilde{S}(U_T, V, N)$$

where  $U_T = U_T(V, N)$  is the unique solution of the equation

$$T_S(U_T(V, N), V, N) = T$$

for fixed values of  $N$  and  $V$ .

Proof: Exercise !!!

Example (5.8): let us use the second version to calculate the free energy for the monatomic ideal gas. Recall,

$$\tilde{S}(U, V, N) = N s_0 + NR \ln \left( \left( \frac{U}{U_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-5/2} \right),$$

where  $s_0, R, U_0, V_0, N_0$  are positive constants.

Then

$$\frac{\partial \tilde{S}}{\partial U} = \frac{NR}{\left( \frac{U}{U_0} \right)^{3/2}} \cdot \frac{3}{2} \left( \frac{U}{U_0} \right)^{1/2} \frac{1}{U_0}$$

$$= \frac{3NR}{2U_0} \cdot \frac{U_0}{U}$$

$$= \frac{3NR}{2U}$$

Thus

$$T_S(U, V, N) = \frac{2}{3} \frac{U}{NR}$$

and

$$U_T = \frac{3NRT}{2}$$

Therefore,

$$\tilde{F}(T, V, N) = \frac{3NRT}{2} - T N s_0 - NRT \ln \left( \left( \frac{3NRT}{2U_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-5/2} \right)$$

Define

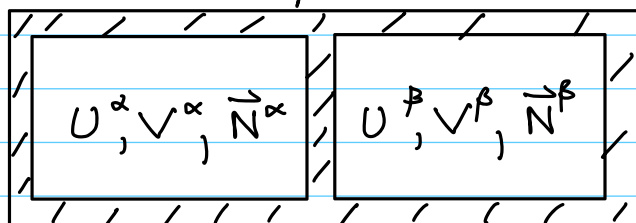
$$T_0 := \frac{2U_0}{3N_0R}$$

Then,

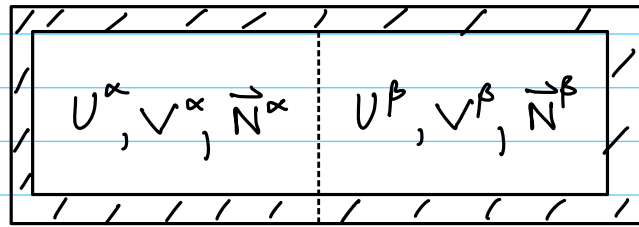
$$\tilde{F}(T, V, N) = \frac{3NRT}{2} - T N s_0 - NRT \ln \left( \left( \frac{T}{T_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-1} \right)$$

### Equilibrium with a Heat Bath

Consider two isolated systems that are initially isolated from each other.



Now, let us remove the isolating wall so that energy, volume, and mass may be exchanged



The entropy will attain its maximum at equilibrium and

$$\begin{aligned} T^\alpha &= T^\beta \\ P^\alpha &= P^\beta \\ \mu_i^\alpha &= \mu_i^\beta, \quad i = 1, \dots, r. \end{aligned}$$

How can we characterize equilibrium?

Now, suppose that instead of isolating from the universe, our composite system is put into contact with a heat bath.

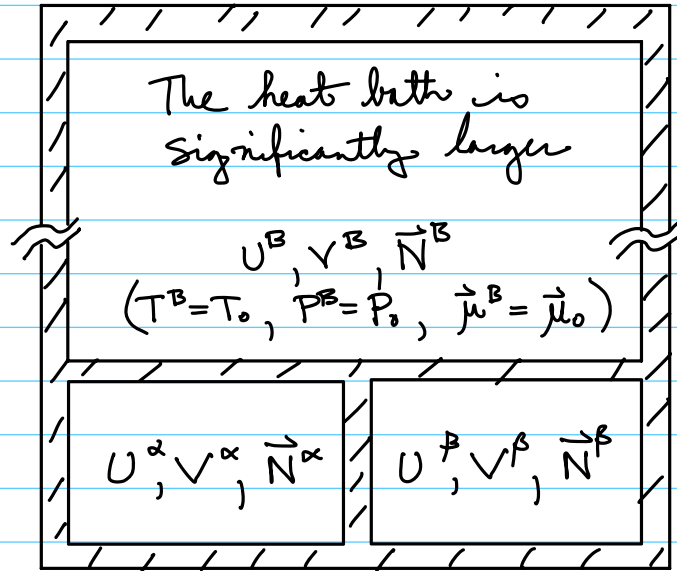
Definition (5.9): A heat bath is an otherwise-isolated thermodynamic system that is so large that, when it exchanges a finite amount of energy with an ordinary, otherwise-isolated, compound system its temperature, pressure, and chemical potential changes are so small as to be negligible.

Similarly, we have

heat-pressure bath: finite exchanges of energy and volume lead to negligible changes in  $T$ ,  $P$ , and  $\mu$  in bath.

A heat-pressure-chemical bath is defined analogously.

let us consider a picture for the case of the heat baths.



let us recall the "integrated form" of the entropy functions.

$$\begin{aligned}\tilde{S}^\square &= \tilde{S}^\square(U^\square, V^\square, \vec{N}^\square) \\ &= \frac{1}{T^\square} U^\square + \frac{P^\square}{T^\square} V^\square - \frac{\vec{\mu}^\square}{T^\square} \cdot \vec{N}^\square\end{aligned}$$

where

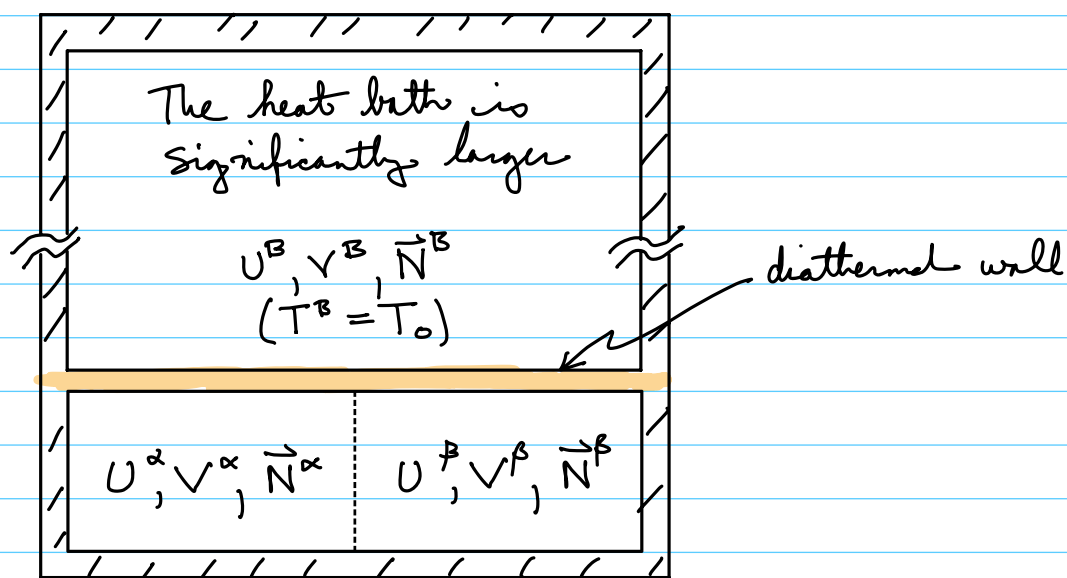
$$T^\square = T^\square(U^\square, V^\square, \vec{N}^\square)$$

et cetera. We will write

$$\begin{aligned}S^{\text{Tot}} &= \tilde{S}^B + \tilde{S}^\alpha + \tilde{S}^\beta \\ &= \tilde{S}^B + \tilde{S}^{\alpha+\beta}\end{aligned}$$

Now, our system will not exchange volume or mass with the baths, only energy, because we will replace the horizontal isolation

wall with a diathermal wall.



We know that, at equilibrium

$$T^\alpha = T^\beta = T^B = T_0.$$

Recall the deviation of  $T^B$  from  $T_0$  at equilibrium is assumed negligible.

Systems  $\alpha, \beta$ , and  $B$  can exchange energy. Systems  $\alpha$  and  $\beta$  can exchange volume and mass with each other but not with the bath.

Thus,

$$\mathcal{S}^{\text{Tot}, \text{eq}} = \max_{\substack{U^\alpha + U^\beta + U^B = U_0 \\ V^\alpha + V^\beta = V_0 \\ N_1^\alpha + N_1^\beta = N_{1,0}}} \left\{ \tilde{S}^B + \tilde{S}^{\alpha+\beta} \right\}$$

Now

$$\tilde{S}^B = \frac{1}{T_0} U^B + \underbrace{\frac{P_0}{T_0} V^B + \frac{\vec{\mu}_0 \cdot \vec{N}^B}{T_0}}_{=\text{const}, G_1}$$

The change in  $T^B, P^B, \mu^B$  from  $T_0, P_0, \mu_0$  is negligible because the bath is so large.

Writing

$$U^B = U_0 - U^\alpha - U^\beta$$

$$\begin{aligned}\tilde{S}^B &= -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + \overbrace{\frac{U_0}{T_0}}^{=\text{const}, C_2} + C_1 \\ &= -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + C_2\end{aligned}$$

Thus,

$$S^{\text{Tot,eq}} = \max_C \left\{ -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + C_2 + \tilde{S}^{\alpha+\beta} \right\}$$

$$= C_2 + \max_C \left\{ -\frac{(U^\alpha + U^\beta)}{T_0} + \tilde{S}^\alpha + \tilde{S}^\beta \right\}$$

$$= C_2 + \max_C \left\{ \frac{-(F^\alpha + F^\beta)}{T_0} \right\}$$

$$= C_2 - \frac{1}{T_0} \min_{\substack{V^\alpha + V^\beta = V_0 \\ N_i^\alpha + N_i^\beta = N_{i,0}}} \left\{ F^\alpha(T_0, V^\alpha, \vec{N}^\alpha) + F^\beta(T_0, V^\beta, \vec{N}^\beta) \right\}$$

Postulate VI: Consider a compound, otherwise-isolated thermodynamic system in contact with an otherwise-isolated heat bath at temperature  $T_0$ . Equilibrium of the compound system is the state satisfying the isothermal condition

$$T^\alpha = T^\beta = T_0$$

which minimizes

$$F^\alpha(T_0, V^\alpha, \vec{N}^\alpha) + F^\beta(T_0, V^\beta, \vec{N}^\beta)$$

subject to the constraints

$$V^\alpha + V^\beta = V_0, \quad (\text{volume conservation})$$

$$N_i^\alpha + N_i^\beta = N_{i,0}. \quad (\text{mass conservation})$$

Remark: Since the system  $\alpha + \beta$  exchanges energy with the heat bath, energy is not conserved in  $\alpha + \beta$ !

Energy is conserved in the system  $B + \alpha + \beta$ .